

→ why the name vector?

The word "vector" comes from the Latin word "vector" which means "carrier".
Also in mathematics and physics, a vector is called that because it carries or shows-

- a magnitude (how much), and.
- a direction (which way).

ex. walk 5 meters to the north.
Here.

- 5 meters = magnitude.
- north = direction

First year

Mathematics:

chapter 3 vectors.

→ What is vector:- is a quantity that has both magnitude (size or amount) and direction.

• Magnitude: This is the length of the vector, representing its size, for example, a car speed.

• Direction: This is the vector's points, for example, North, down, or at 45-degree angle.

→ Geometrically, a vector is typically drawn as an arrow where:

1. The length of the arrow shows the magnitude.
ex. —

2. The arrowhead shows the direction.

→ A simple example
vector is

velocity (speed and direction)
Force (a push or pull)

Magnitude (size)	Direction
60 ^{km} per hour	Due East
100 Newton	straight down

→ Scalar Quantity: In contrast, have only magnitude, like speed 60 (mph), Temp (25°C) or mass (kg)

or
→ The quantity that is completely specified only by its magnitude with an appropriate unit, is called scalar quantity or scalar.

ex:

mass, volume, distance.

3.1.2.

⇒

Vector in a plane (or in coordinate plane)
When we say a vector in a plane, we mean a vector that lies inside a 2D coordinate plane - usually the x-y plane.

so a vector in the plane is simply a quantity that,

- starts from one point (the tail)
- ends at another point (the head)
- and has both magnitude and direction all within the same 2D plane.

ex:

$$\vec{a} = (3, 4)$$

That means:

- Move 3 units along the x-axis
- Move 4 units along the y-axis

⇒ Vector Representation:

Two types:

1. Symbolic Representation →

2. Graphical Representation.

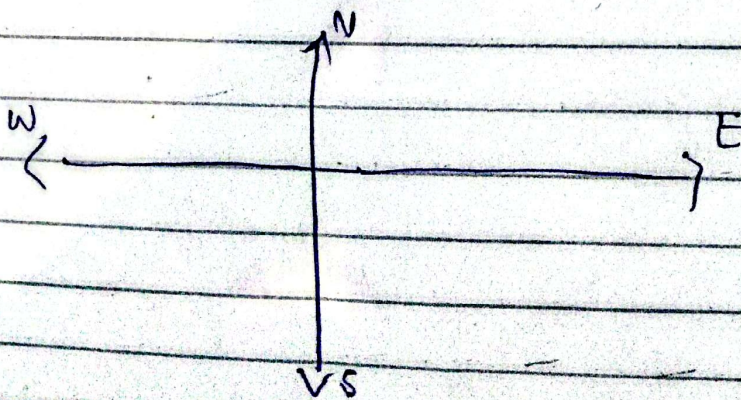
Def:

Graphically a vector can be represented by Directed line segment, whose length is determined by the magnitude of given vector ^{Tukao (piece)} according to a chosen scale.

1. Line means simple line, who doesn't have any curve.

2. Directed means it must have directed line.

Ex: velocity which is 30 metre per East.



• A vector is just a quantity that has both size (magnitude) and direction. When you draw it on paper (graphically), you use an arrow (a directed line segment) to show it.

1, The arrowhead: This shows the direction the vector is pointing (eg North up or decrease to the right)

2, The length of the arrow: This shows the size or magnitude of the vector (eg, the speed, the force, or the distance)

The role of scale.

Because you can't always draw an arrow that is exactly as long as the real-world magnitude (like drawing an arrow 500 miles long!) you have to use a scale.

→ scale means you decide that a certain length on your paper will represent a certain value in the real world.

ex: Representing a car velocity
a car traveling at a speed of 50 miles per hour (magnitude) East (direction).
(This is a velocity vector)
→ you can't draw an arrow 50 miles long, so you choose a scale.